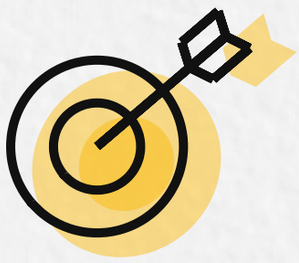
Week 6 - Learning Map



Goal:

Build a reliable real-estate negotiation system that uses FSMs for safe loops, MCP for tool access, LangGraph for orchestration, and Google ADK with A2A for structured agent communication across multiple rounds, services, and negotiation states.

Overview...

Week 6 focuses on building reliable multi-agent systems through a real-estate negotiation simulator where buyer and seller agents interact under clear rules and bounded rounds.

The module starts with the problem of fragile agent systems that rely on raw strings, weak parsing, and uncontrolled loops, then introduces finite state machines to enforce valid transitions and guaranteed termination.

From there, learners explore Model Context Protocol, which allows agents to discover and call structured tools instead of depending only on prompt context.

This is applied to negotiation by fetching live market and pricing information through external tool servers.

The module then moves into LangGraph, where the negotiation becomes a typed, stateful workflow with explicit nodes, edges, reducers, and shared memory across turns.

Finally, you will use Google ADK and the A2A protocol to expose agents as networked services that communicate through structured task messages.

Together, the repo and live class show how to evolve from fragile demos into production-style agent systems with safe loops, tools, orchestration, and agent-to-agent communication.

Note:

Post-class resources are optional and intended for post-class learning.

HOW TO START?

Roadmap

Pre-class Setup

- Clone the project repo
- Install required dependencies
- Set up OpenAI API key and environment variables

Learners are encouraged to follow along with the instructor during the live session and execute the code on their local environment.

Live Class

Reliable Agent Foundations

- Naive negotiation vs FSM-based design
- Why bounded states and transitions matter
- Build safe, terminating agent workflows

Tool Use + Multi-Agent Orchestration

- MCP for structured tool access
- LangGraph for stateful agent workflows
- Buyer and seller agents share negotiation state across multiple rounds

Production-Style Agent Communication

- Google ADK abstractions
- A2A protocol for agent-to-agent messaging
- Build the real-estate negotiation system end to end

Post-class resources (Optional deep dives)

MCP, A2A, & ADK

Assignment

Please use the Wednesday Technical Coaching session to clarify any doubts you have on the topics. The assignment solutions will be discussed during the Thursday Assignment Review session.

All the resources have been added to your Uplevel account. If you have any questions, please reach out to the Operations team or raise a support ticket.